Type MS · Arithmetic: standard form, estimates and rounding

Answer 1 of 7 · Question 5 part f · 1 mark

2024 JUN · Paper 02 · extract 1 of 3 · find part f in this mark scheme extract

Question number	Answer	Mark
5(a)	**One** mark for correct knowledge of a case study. • A case study is in-depth research often undertaken with a single person that collects a range of data using different methods (1). Accept any other appropriate response.	(1)

Question number	Answer	Mark
5(b)	**One** mark for correct knowledge of quantitative data. **One** mark for an appropriate example from the scenario. • Quantitative data is data that can be counted or measured numerically (1), such as measuring the alcohol consumed by the patient each week in units (1). Accept any other appropriate response.	(2)

Question number	Answer	Mark
5(c)	**One** mark for an appropriate estimation. 300 - 60 • 240 Accept any other appropriate response.	(1)

Question number	Answer	Mark
5(d)	**One** mark for correct calculation of the mode for units of alcohol consumed. • 79 (1). Reject all other answers.	(1)

Type MS · Arithmetic: standard form, estimates and rounding

Answer 1 of 7 · Question 5 part f · 1 mark

2024 JUN · Paper 02 · extract 2 of 3 · find part f in this mark scheme extract

Question number	Answer	Mark
5(e)	**One** mark for correct rank ordering. **One** mark for correct answer. • Rank ordering: 34 45 63 70 72 77 79 79 • (70+72)/2 = 71 Accept any other appropriate response.	(2)

Question number	Answer	Mark
5(f)	**One** mark for accurately giving 12.81924 to two decimal places. For example; • 12.82 (1). Reject all other answers.	(1)

Type MS · Arithmetic: standard form, estimates and rounding

Answer 1 of 7 · Question 5 part f · 1 mark

2024 JUN · Paper 02 · extract 3 of 3 · find part f in this mark scheme extract

Type MS · Arithmetic: standard form, estimates and rounding

Answer 2 of 7 · Question 3 part a · 1 mark

2019 JUN · Paper 02 · extract 1 of 2 · find part a in this mark scheme extract

Question number	Answer	Mark
3(a)	**One** mark for correct calculation. • 9-5 = 4 Reject all other answers.	(1)

Question number	Answer	Mark
3(b)	**One** mark for correct answer. Calculation: 12:18 simplified using a common factor • 12/6:18/6 = 2:3 Accept any other appropriate response.	(1)

Type MS · Arithmetic: standard form, estimates and rounding

Answer 2 of 7 · Question 3 part a · 1 mark

2019 JUN · Paper 02 · extract 2 of 2 · find part a in this mark scheme extract

Question number	Answer	Mark
3(c)	**One** mark for identification of a relevant improvement **One** mark for justification, which must be linked to the improvement identified For example: • Vanessa could observe driving behaviour on a number of different roads in different towns (1). This would improve the representativeness of the findings about driving speed of young and mature drivers in general (1). Accept any other appropriate response.	**(2)**

Type MS · Arithmetic: standard form, estimates and rounding

Answer 3 of 7 · Question 6 part a · 1 mark

2019 JUN · Paper 02 · extract 1 of 3 · find part a in this mark scheme extract

Question number	Answer	Mark
6(a)	**One** mark for a correct answer • $39,980,000 = 3.998 \times 10^7$ Reject all other answers.	**(1)**

Type MS · Arithmetic: standard form, estimates and rounding

Answer 3 of 7 · Question 6 part a · 1 mark

2019 JUN · Paper 02 · extract 2 of 3 · find part a in this mark scheme extract

Question number	Answer	Mark
6(b)	**One** mark for an appropriate estimation 40,000,000 – 10,000,000 • 30,000,000 Accept any other appropriate response.	(1)

Question number	Answer	Mark
6(c)	**One** mark for a correct answer Calculation: 63% = 0.63 0.63 x 39,980,000 = 25,187,400 • 25,187,400 Reject all other answers.	(1)

Question number	Answer	Mark
6(d)	**One** mark for correct identification of primary data example **One** mark for correct identification of secondary data example **Type of data** / **Example from Kayla's investigation** Primary data — **Online questionnaire** Secondary data — **Two newspaper articles** Accept any other appropriate response.	(2)

Type MS · Arithmetic: standard form, estimates and rounding

Answer 3 of 7 · Question 6 part a · 1 mark

2019 JUN · Paper 02 · extract 3 of 3 · find part a in this mark scheme extract

Type MS · Arithmetic: standard form, estimates and rounding

Answer 4 of 7 · Question 6 part b · 1 mark

2019 JUN · Paper 02 · extract 1 of 3 · find part b in this mark scheme extract

Question number	Answer	Mark
6(a)	**One** mark for a correct answer • 39,980,000 = 3.998×10^7 Reject all other answers.	(1)

Type MS · Arithmetic: standard form, estimates and rounding

Answer 4 of 7 · Question 6 part b · 1 mark

2019 JUN · Paper 02 · extract 2 of 3 · find part b in this mark scheme extract

Question number	Answer	Mark
6(b)	**One** mark for an appropriate estimation 40,000,000 – 10,000,000 • 30,000,000 Accept any other appropriate response.	(1)

Question number	Answer	Mark
6(c)	**One** mark for a correct answer Calculation: 63% = 0.63 0.63 x 39,980,000 = 25,187,400 • 25,187,400 Reject all other answers.	(1)

Question number	Answer	Mark
6(d)	**One** mark for correct identification of primary data example **One** mark for correct identification of secondary data example **Type of data** / **Example from Kayla's investigation** Primary data — **Online questionnaire** Secondary data — **Two newspaper articles** Accept any other appropriate response.	(2)

Type MS · Arithmetic: standard form, estimates and rounding

Answer 4 of 7 · Question 6 part b · 1 mark

2019 JUN · Paper 02 · extract 3 of 3 · find part b in this mark scheme extract

Type MS · Arithmetic: standard form, estimates and rounding

Answer 5 of 7 · Question 3 part e · 2 marks

2020 NOV · Paper 02 · extract 1 of 3 · find part e in this mark scheme extract

Question number	Answer	Mark
3(a)	**One** mark for correct knowledge of a feature of a laboratory experiment For example: • A laboratory experiment takes place in an artificial environment for the participants (1). Accept any other appropriate response.	(1)

Question number	Answer	Mark
3(b)	**One** mark for knowledge of sampling method **One** mark for application to the scenario For example: • A volunteer sample is where people offer to take part in a piece of research (1), so Becca could advertise her experiment and ask for males and females to sign up to take part (1). Accept any other appropriate response	(2)

Type MS · Arithmetic: standard form, estimates and rounding

Answer 5 of 7 · Question 3 part e · 2 marks

2020 NOV · Paper 02 · extract 2 of 3 · find part e in this mark scheme extract

Question number	Answer	Mark
3(c)	**One** mark for knowledge of single-blind technique. **One** mark for application to the scenario. • A single-blind technique hides the true aim of the study from the participants taking part (1) so that the males do not change their behaviour and try harder on the coordination task (1). Accept any other appropriate response.	(2)

Question number	Answer	Mark
3(d)	**One** mark for correct answer. Calculation: 30:25 simplified using common factor 30/5:25/5 • 6:5 Reject all other answers.	(1)

Question number	Answer	Mark
3(e)	**One** mark for calculating the difference. **One** mark for answer to two significant figures. • 63.51-57.24=6.27 (1) • 6.3 (1) Accept any other appropriate response.	(2)

Question number	Answer	Mark
3(f)	**One** mark for conclusion made. **One** mark for justification of conclusion through analysis of data. • Females are better at word-meaning tasks than males (1) as the female participants were 12.15 seconds quicker than the male participants at this task (1). Accept any other appropriate response.	(2)

Type MS · Arithmetic: standard form, estimates and rounding

Answer 5 of 7 · Question 3 part e · 2 marks

2020 NOV · Paper 02 · extract 3 of 3 · find part e in this mark scheme extract

Type MS · Arithmetic: standard form, estimates and rounding

Answer 6 of 7 · Question 3 part b · 1 mark

2021 NOV · Paper 02 · find part b in this mark scheme extract

Question number	Answer	Mark
3(a)	**One** mark for calculating the mean. Calculation: 458+211+641+189=1499/4 • 374.75 Accept any other appropriate response.	(1)

Question number	Answer	Mark
3(b)	**One** mark for a correct calculation. 458+211+641+189+931+439+838+271 • 3978 Reject all other answers.	(1)

Question number	Answer	Mark
3(c)	**One** mark for correct calculation. Calculation: 931-271 • 660 Accept any other appropriate response.	(1)

Type MS · Arithmetic: standard form, estimates and rounding

Answer 7 of 7 · Question 5 part c · 1 mark

2024 JUN · Paper 02 · extract 1 of 3 · find part c in this mark scheme extract

Question number	Answer	Mark
5(a)	**One** mark for correct knowledge of a case study. • A case study is in-depth research often undertaken with a single person that collects a range of data using different methods (1). Accept any other appropriate response.	(1)

Question number	Answer	Mark
5(b)	**One** mark for correct knowledge of quantitative data. **One** mark for an appropriate example from the scenario. • Quantitative data is data that can be counted or measured numerically (1), such as measuring the alcohol consumed by the patient each week in units (1). Accept any other appropriate response.	(2)

Question number	Answer	Mark
5(c)	**One** mark for an appropriate estimation. 300 - 60 • 240 Accept any other appropriate response.	(1)

Question number	Answer	Mark
5(d)	**One** mark for correct calculation of the mode for units of alcohol consumed. • 79 (1). Reject all other answers.	(1)

Type MS · Arithmetic: standard form, estimates and rounding

Answer 7 of 7 · Question 5 part c · 1 mark

2024 JUN · Paper 02 · extract 2 of 3 · find part c in this mark scheme extract

Question number	Answer	Mark
5(e)	**One** mark for correct rank ordering. **One** mark for correct answer. • Rank ordering: 34 45 63 70 72 77 79 79 • (70+72)/2 = 71 Accept any other appropriate response.	(2)

Question number	Answer	Mark
5(f)	**One** mark for accurately giving 12.81924 to two decimal places. For example; • 12.82 (1). Reject all other answers.	(1)

Type MS · Arithmetic: standard form, estimates and rounding

Answer 7 of 7 · Question 5 part c · 1 mark

2024 JUN · Paper 02 · extract 3 of 3 · find part c in this mark scheme extract